

Fiber-Optic Gyroscope Signal Denoising Using an Adaptive Robust Kalman Filter

Mundla Narasimhappa, Samrat L. Sabat, and Jagannath Nayak

Abstract—Random noise is an important issue in interferometric fiber-optic gyroscope (IFOG). In this paper, an adaptive robust Kalman filter (KF) and a variant of this are applied to minimize the random noise in IFOG. In the variant of the adaptive robust KF, the measurement noise covariance matrix is adapted using the weighted covariance of the innovation sequence. The suitability of both the algorithms is studied for denoising the IFOG signal under static and maneuvering conditions. In the static case, the Allan variance analysis and the conventional variance are used as the performance indicators to determine the efficiency of the algorithm. In the maneuvering case, root mean-squared error is considered as the performance indicator. The performance of both the algorithms is compared with the conventional KF, innovation-based adaptive estimation adaptive KF, and for minimizing random noise. The experimental results reveal that both the algorithms are competitive for denoising the IFOG signal.

Index Terms—Strap down inertial navigation system (SINS), interferometric fiber-optic gyro (IFOG), gyro drift, adaptive Kalman filter, conventional KF, Allan variance.

I. INTRODUCTION

THE integration of strap down inertial navigation system (SINS) and global position navigation system is a promising technology for providing accurate state information (like position and attitude) of a moving object. The interferometric fiber optic gyroscope (IFOG) is used in SINS to measure the angular rotation rate of an object [1]. The working principle of IFOG is based on the Sagnac effect, and it has advantages such as wide dynamic range, long life, short start-up time and simple structure, over Ring laser gyro (RLG) and MEMS gyros. Due to these features, IFOGs are being used for military and defense applications [2].

The precision of IFOG output degrades due to random noises. An imperfect fabrication, fluctuations in the ambient temperature and pressure during the operation of the device as well as electronics of the IFOG contributes to this noise [3], [4]. Allan variance method is a popular time domain method to identify the source and quantify the random noise

in an IFOG [5]. Random noises are mainly quantization noise (Q), bias instability (Bs), angle random walk (N), rate random walk (K) and rate ramp (R) [6].

Different methods like discrete wavelet transforms (DWT) and empirical mode decomposition (EMD) method are used for eliminating the random noise of IFOG signal [7]–[10]. It is exceptionally difficult to compensate the random noise [11], [12]. In the maneuvering case, the IFOG noise characteristic is more complex and time-varying. Due to this, the above-mentioned methods fail to eliminate the random noise efficiently [13]–[15]. For minimizing random noise the system to be modeled accurately. Conventional Kalman filter (CKF) is widely used technique for estimating the system data output with given noisy measurement data. It is used for minimizing the additive white noise (angle random walk) of gyroscope [16]–[18]. Hybrid Kalman filter techniques are used to improve the accuracy of IFOG by minimizing random noise in FOG sensor [19], [20].

The accuracy of conventional KF depends on the value of process and measurement noise covariance matrices. Tuning an appropriate value for these parameters for real time IFOG application is a crucial issue. An improper value of these matrices lead to filter divergence. Adaptive Kalman filter (AKF) overcomes this by adaptively changing noise parameters of the model. AKF is useful when system models are partially unknown and possibly time-varying [21]. In AKF, the process noise and measurement noise covariance matrices are adaptively estimated based on either innovation sequence adaptive estimation (IAE) or residual sequence adaptive estimation (RAE) or multiple model adaptive estimations (MMAE). In RAE and IAE Kalman filter, the innovation or residual sequence must be known a priori [21]. IAE is a popular technique among the adaptive estimation techniques. In this method, the innovation sequence is used to estimate the noise covariance matrices using adaptive methods like Bayesian, maximum likelihood, correlation and covariance matching. In covariance matching technique, the noise covariance matrix is estimated by window average method [22]. The application of IAE-AKF and RAE-AKF is explored in inertial navigation system [23]. In recent literature, AKF algorithms are used to minimize random gyro noise for improving the accuracy of gyro sensor [24]. The Kalman filter accuracy depends on the accuracy of the state model, the estimated values of noise parameters and the learning strategy used for adapting the noise parameters. In IAE-AKF although noise parameters (Q and R) are estimated adaptively but the adaptive rate is independent of learning statistics. This leads slow adaption rate. In the maneuvering case, IFOG signal changes

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rapidly. To minimize the noise of a rapidly changing signal a Kalman filter with fast adaptation rate is required. In the literature, adaptive robust Kalman filter is proposed based on the robust estimation theory with learning statistics.

Robust estimation methods have been developed based on min-max (M-M) theory to suppress the influence of observation errors from the Gaussian distribution data [25], [26]. The maximum likelihood estimation (MLE) and Bayesian estimation techniques have been developed for a robust filter to suppress the influence of process model and measurement model errors [27]. In M-M filter, the measurement model errors and the process noise statistics are resisted by equivalent weighted function, and this helps to overcome the divergent effect [28], [29]. Adaptive robust Kalman filter (ARKF) is used to suppress the influence of outliers in the state model [30]. In ARKF, segment function and learning statistics are introduced to suppress the influence of residual and to minimize the effect of outliers in measurement [27]. In real time, sensor noise may produce outlier in the observation. The ARKF algorithms have been applied in navigation applications and also used to reduce the white and colored noises [31], however, its suitability for minimizing random noise in IFOG is not explored so far.

This paper presents a variant of ARKF to minimize the random noise of IFOG signal. In the modified algorithm, data samples are associated with an individual weight indicating its contribution to the state estimate at each epoch. The weights are calculated based on the estimated variance of measurement signal values. In this method, the disturbances of predicted state error model due to outlier measurement is controlled by an adaptive factor obtained using three segment approach. The modified algorithm is named as weighted covariance adaptive robust Kalman filter (WCARKF). In both ARKF and WCARKF the R adaption is based on the learning statistics. Thus, it is expected that these algorithms will be able to minimize random noise in case of both static and dynamic IFOG signal. In this paper, the suitability of both ARKF and WCARKF for minimizing random noise in IFOG signal is studied.

The rest of this paper is organized as follows: Section II analyzes the IFOG noise model. Section III presents CKF and IAE-AKF algorithm. Section IV describes adaptive Kalman filter based on Robust estimation theory (ARKF). Section V presents the proposed algorithm for minimizing random noise in IFOG FOG signal. Section VI shows the experimental results followed by conclusions in section VII.

II. IFOG NOISE SOURCE ANALYSIS

In IFOG, the random errors are composed of short term and long term errors. Short term errors are high-frequency components, whereas the long-term errors are low-frequency components. High-frequency components are characterized by white noise whereas low frequency components are characterized by $1/f$ noise [4]. The IFOG output can be expressed as

$$\omega_{out} = \omega_k + \omega_{ek} + b_k + e_k \quad (1)$$

where ω_k is the actual input angular velocity, ω_{ek} is Earth rotation rate component on gyro sensitive axis, b_k is gyro zero bias, e_k is the random noise of IFOG. This paper only focus on analyzing FOG random noise and propose an algorithm to eliminate/minimize the random noise. Allan variance analysis method is used to identify and quantify the noise present in the signal [5]. In the signal the sum of each noise terms is

$$\sigma_{Total}^2(\tau) = \sigma_Q^2(\tau) + \sigma_N^2(\tau) + \sigma_{Bs}^2(\tau) + \sigma_K^2(\tau) + \sigma_R^2(\tau) \quad (2)$$

The output of IFOG includes quantization noise (Q), angle rate random walk (N), bias instability (Bs), rate random walk (K), rate ramp (RR) and exponentially correlated noise [5], [6].

The angle random walk noise and bias instability noise models can be expressed as [13], [32]

$$\mathbf{e}_k = \mathbf{e}_{k-1} + \Delta T(\mathbf{b}_{k-1}) \quad (3)$$

$$\mathbf{b}_k = \mathbf{b}_{k-1} \quad (4)$$

where $\mathbf{e}_k$ is the angle random walk that follows normal distribution with zero mean and finite variance $\mathbf{Q}_k$, $\mathbf{b}_k$ is the gyro bias coupled with white noise (Γ_k) and ΔT is the sampling time. In this model, the control input is zero whereas, $\mathbf{e}_k$ and $\mathbf{b}_k$ follows $E(e_k) = 0$ and $E(\mathbf{b}_k) = 0$. These noise models can be expressed using state space model as

$$\begin{aligned} \mathbf{x}_k &= \begin{bmatrix} \mathbf{e}_k \\ \mathbf{b}_k \end{bmatrix} = \begin{bmatrix} 1 & \Delta T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{e}_{k-1} \\ \mathbf{b}_{k-1} \end{bmatrix} + \mathbf{w}_k \\ \mathbf{z}_k &= \begin{bmatrix} 1 & 0 \end{bmatrix} \mathbf{x}_k + \mathbf{v}_k \end{aligned} \quad (5)$$

where $\mathbf{v}_k$, $\mathbf{w}_k$, $\mathbf{z}_k$ and $\mathbf{x}_k$ are the measurement noise, process noise, measurements and state vector respectively.

The noise statistics follow

$$E \left\{ \begin{bmatrix} \mathbf{w}_k \\ \mathbf{v}_k \end{bmatrix} \begin{bmatrix} \mathbf{w}_l & \mathbf{v}_l \end{bmatrix} \right\} = \begin{bmatrix} \sigma_N^2(\tau) & 0 \\ 0 & \sigma_{Bs}^2(\tau) \end{bmatrix} \delta_{kl} \quad (6)$$

where, δ_{kl} is Kronecker delta function i.e.,

$$\delta_{kl} = \begin{cases} 1, & k = l. \\ 0, & \text{otherwise.} \end{cases}$$

In this paper, the above state space model is used for analysis.

III. CONVENTIONAL AND ADAPTIVE KALMAN FILTERING

A. Conventional Kalman Filter

The application of conventional Kalman filter for IFOG requires a prior knowledge of dynamic process and measurement models, in addition to the process and measurement noise of the IFOG. Considering a linear dynamic system, the state and measurement equations can be written as

$$\mathbf{x}_k = \mathbf{A}\mathbf{x}_{k-1} + \mathbf{w}_{k-1} \quad (7)$$

$$\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k \quad (8)$$

where, $\mathbf{x}_k$ and $\mathbf{A}$ are the state vector and state transition matrix at epoch k respectively, $\mathbf{w}_{k-1}$ is the system noise; $\mathbf{z}_k$ is the observation at epoch k ; $\mathbf{H}$ represents the observation matrix, $\mathbf{v}_k$ is the measurement noise. Let us assume that the process and measurement noises $\mathbf{w}_{k-1}$, $\mathbf{v}_k$ are white Gaussian zero mean

noise and statistically independent from each other, following the below properties

$$E[\mathbf{w}_{k-1}] = \mathbf{0} \quad (9)$$

$$E[\mathbf{v}_k] = \mathbf{0} \quad (10)$$

$$E[\mathbf{w}_k \mathbf{w}_l^T] = \mathbf{Q}_k \delta_{kl} \quad (11)$$

$$E[\mathbf{v}_k \mathbf{v}_l^T] = \mathbf{R}_k \delta_{kl} \quad (12)$$

where, $\mathbf{Q}_k$ and $\mathbf{R}_k$ are the process and measurement noise covariance matrices, respectively. δ_{kl} is Kronecker delta function i.e.,

$$\delta_{kl} = \begin{cases} 1, & k = l. \\ 0, & k \neq l. \end{cases}$$

The prediction equations can be expressed as

$$\hat{\mathbf{x}}_k^- = \mathbf{A} \hat{\mathbf{x}}_{k-1} \quad (13)$$

$$\hat{\mathbf{P}}_k^- = \mathbf{A} \hat{\mathbf{P}}_{k-1} \mathbf{A}^T + \mathbf{Q}_k \quad (14)$$

In the above equations, $\mathbf{A}$ is the state transition matrix and $\mathbf{A}^T$ denotes the transpose of $\mathbf{A}$. $\hat{\mathbf{P}}_k^-$ and $\mathbf{Q}_k$ represents prediction state error covariance and process noise covariance matrix at epoch k .

In the linear Kalman filter, the measurement updated equations are

$$\mathbf{K}_k = \hat{\mathbf{P}}_k^- \mathbf{H}^T [\mathbf{H} \hat{\mathbf{P}}_k^- \mathbf{H}^T + \mathbf{R}_k]^{-1} \quad (15)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k [\mathbf{z}_k - \mathbf{H} \hat{\mathbf{x}}_k^-] \quad (16)$$

$$\hat{\mathbf{P}}_k = [\mathbf{I} - \mathbf{K}_k \mathbf{H}] \hat{\mathbf{P}}_k^- \quad (17)$$

The initial state and error covariance matrix of the state are assumed to be $\hat{\mathbf{x}}_0$ and $\hat{\mathbf{P}}_0$ respectively. The $\mathbf{K}_k$ is the Kalman gain. $\mathbf{R}_k$ and $\mathbf{I}$ are the measurement noise covariance matrix and identify matrix respectively [23].

B. Innovation Based Adaptive Estimation Adaptive Kalman Filtering (IAE-AKF)

In this algorithm, the innovation sequence is assumed to represent the additional information available to the filter as a consequence of the new observation $\mathbf{z}_k$ [23]. The innovation sequence $\mathbf{V}_k$ is a zero-mean white Gaussian noise sequence. The innovation sequence can be written as [21]

$$\mathbf{V}_k = \mathbf{z}_k - \mathbf{H} \hat{\mathbf{x}}_k^- \quad (18)$$

The weighted innovation $\mathbf{K}_k [\mathbf{z}_k - \mathbf{H} \hat{\mathbf{x}}_k^-]$, acts as a correction to the predicted estimation $\hat{\mathbf{x}}_k^-$ to form $\hat{\mathbf{x}}_k$. By substituting the measurement model (8) in (18), we get $\mathbf{V}_k = \mathbf{H}(\mathbf{x}_k - \hat{\mathbf{x}}_k^-) + \mathbf{v}_k$. By taking variance on both sides of this, the theoretical covariance matrix of $\mathbf{V}_k$ is [23]

$$\mathbf{C}_{V_k} = \mathbf{H} \hat{\mathbf{P}}_k^- \mathbf{H}^T + \mathbf{R}_k \quad (19)$$

The optimal estimation of covariance matrix of innovation sequence using average window method can be expressed as [33]

$$\hat{\mathbf{C}}_{V_k} = \frac{1}{D_s} \sum_{j=j_0}^k \mathbf{V}_j \mathbf{V}_j^T \quad (20)$$

where $\mathbf{V}_j$ is the innovation sequence. D_s is the window size, $j_0 = k - D_s + 1$ is the first epoch. If the window size is too small, the measurement estimation covariance can be noisy, on the other hand, the estimation of measurement covariance will be smoother. Usually, window size is chosen empirically for statistical smoothing [33].

The estimated measurement noise covariance based on innovation sequence is

$$\hat{\mathbf{R}}_k = \hat{\mathbf{C}}_{V_k} - \mathbf{H} \hat{\mathbf{P}}_k^- \mathbf{H}^T \quad (21)$$

where, $\hat{\mathbf{R}}_k$ is the estimated measurement noise covariance matrix, $\mathbf{H}$ is the observation matrix, $\hat{\mathbf{P}}_k^-$ is the prediction state error covariance and $\hat{\mathbf{C}}_{V_k}$ is the estimated covariance matrix of innovation sequence.

IV. ADAPTIVE KALMAN FILTER BASED ON ROBUST ESTIMATION THEORY

In the adaptive robust Kalman filter (ARKF), the measurement update equations can be written as [29], [34]

$$\bar{\mathbf{K}}_k = \frac{1}{\alpha_k} \hat{\mathbf{P}}_k^- \mathbf{H}^T \left[\frac{1}{\alpha_k} \mathbf{H} \hat{\mathbf{P}}_k^- \mathbf{H}^T + \hat{\mathbf{R}}_k \right]^{-1} \quad (22)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \bar{\mathbf{K}}_k [\mathbf{z}_k - \mathbf{H} \hat{\mathbf{x}}_k^-] \quad (23)$$

$$\hat{\mathbf{P}}_k = [\mathbf{I} - \bar{\mathbf{K}}_k \mathbf{H}] \hat{\mathbf{P}}_k^- \alpha_k^{-1} \quad (24)$$

where, $\bar{\mathbf{K}}_k$ is adaptive robust Kalman gain, $\hat{\mathbf{x}}_k$ is the estimated state and $\hat{\mathbf{P}}_k$ is estimated state noise covariance. More details about the adaptive robust Kalman filter can be found in [29] and [34].

V. IMPROVED ROBUST ADAPTIVE KALMAN FILTER

In WCARKF, data samples are associated with an individual weight indicating its contribution to the state estimate at each epoch. The weights are calculated based on the estimated variance of measurement state. In WCARKF an adaptive tuning parameter is used to control the disturbances due to outlier. The tuning parameter is evaluated using three segment approach to control the disturbances [35]–[38].

The innovation sequence can be expressed as [34]

$$\bar{\mathbf{V}}_k = \mathbf{A} \hat{\mathbf{x}}_k^- - \mathbf{z}_k \quad (25)$$

In WCARKF, measurement noise covariance matrix $\hat{\mathbf{R}}_k$ is evaluated by using weighted covariance matrix of innovation sequence as

$$\hat{\mathbf{C}}_{V_k} = \frac{1}{D_s} \sum_{j=j_0}^k \omega(\sigma_{k-j}^2) \bar{\mathbf{V}}_j \bar{\mathbf{V}}_j^T \quad (26)$$

where $\omega(\sigma_{k-j}^2)$ and $\bar{\mathbf{V}}_j$ are weight function and innovation sequence respectively.

A. Calculation of Weight function

The estimated variance of measurement state is defined as [36], [37]

$$\hat{\sigma}_{k-j}^2 = \frac{\bar{\mathbf{V}}_k^T \mathbf{R}_k \bar{\mathbf{V}}_k}{\mathbf{r}_k} \quad (27)$$

where, $\tilde{\mathbf{V}}_k$ is the innovation sequence of measurements, $\mathbf{R}_k$ is the measurement noise covariance matrix and r_k denotes the number of measurements [35]–[37]. The adaptive weight function is the estimated variance of measurements defined as

$$\omega(\hat{\sigma}_{k-j}^2) = \left(\frac{1}{\hat{\sigma}_{k-j}^2} \right) / \left(\sum_{j=1}^m \frac{1}{\hat{\sigma}_{k-j}^2} \right) \quad (28)$$

The sum of the weights of measurement are equal to 1 i.e., as

$$\sum_{j=1}^m \omega(\hat{\sigma}_{k-j}^2) = 1 \quad (29)$$

B. Three Segment Method and Learning Statistics Approaches

In the WCARKF technique, three segment method is used to find the optimal adaptive factor such that the theoretical value of predicted measurement is nearly equal to the actual estimated output of the filter [26], [29]. The adaptive factor is used to control the disturbances of predicted state error model and to resist the outlier of measurement in robust estimation [38]. Adaptive factor (α_k) is represented as

$$\alpha_k = \begin{cases} 1, & |\Delta\tilde{\mathbf{X}}_0| \leq c_0. \\ \frac{c_0}{|\Delta\tilde{\mathbf{X}}_k|} \left(\frac{c_1 - |\Delta\tilde{\mathbf{X}}_k|}{|\Delta\tilde{\mathbf{X}}_k|} \right), & c_0 \leq |\Delta\tilde{\mathbf{X}}_0| \leq c_1. \\ 0, & |\Delta\tilde{\mathbf{X}}_k| \geq c_1. \end{cases} \quad (30)$$

where, $\Delta\tilde{\mathbf{X}}_k = \hat{\mathbf{x}}_k - \hat{\mathbf{x}}_k^-$ is the residual vector of the predicted state vector and c_0 and c_1 are two criterion constants ranges between 1.0-1.5 and 3.0-4.5, respectively. A learning statistic of the kinematic model error $|\Delta\tilde{\mathbf{X}}_k|$ is expressed as [29]

$$|\Delta\tilde{\mathbf{X}}_k| = \frac{\|\tilde{\mathbf{V}}_k\|}{\sqrt{\mathbf{C}_{V_k}}} \quad (31)$$

where $\|\tilde{\mathbf{V}}_k\|$ is the norm of innovation vector and $\mathbf{C}_{V_k}$ is the covariance matrix of innovation sequence.

C. Adaption of Measurement Noise Covariance Matrix

In ARKF, the process and measurement noise covariance matrix is used to control the disturbances. The noise covariance is estimated using the innovation sequences and covariance-matching techniques [22], [33]. In WCARKF technique, the innovation based on the weighted window function is used to determine the estimated measurement noise covariance matrix ($\hat{\mathbf{R}}_k$) as

$$\hat{\mathbf{R}}_k = \frac{1}{D_s} \sum_{j=j_0}^k \omega(\sigma_{k-j}^2) \tilde{\mathbf{V}}_j \tilde{\mathbf{V}}_j^T + \mathbf{H} \hat{\mathbf{P}}_k^- \mathbf{H}^T \quad (32)$$

where, $\omega(\sigma_{k-j}^2)$ is the weighted function of measurement error, D_s is the window size, $j_0 = k - D_s + 1$ is the first epoch.

The prediction and measurement updated equations can be written as

$$\hat{\mathbf{x}}_k^- = \mathbf{A} \hat{\mathbf{x}}_{k-1} \quad (33)$$

$$\hat{\mathbf{P}}_k^- = \mathbf{A} \hat{\mathbf{P}}_{k-1}^- \mathbf{A}^T + \mathbf{Q}_k \quad (34)$$

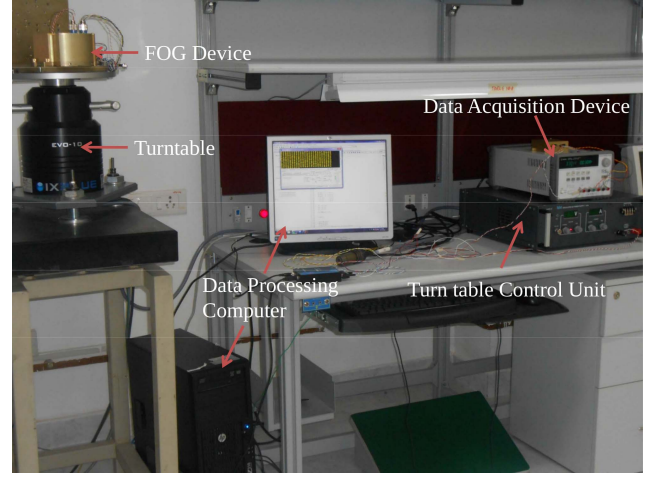


Fig. 1. Experimental setup arrangement of IFOG sensor.

and

$$\bar{\mathbf{K}}_k = \frac{1}{\alpha_k} \hat{\mathbf{P}}_k^- \mathbf{H}^T \left[\frac{1}{\alpha_k} \mathbf{H} \hat{\mathbf{P}}_k^- \mathbf{H}^T + \hat{\mathbf{R}}_k \right]^{-1} \quad (35)$$

$$\hat{\mathbf{x}}_k = (\mathbf{A}^T \hat{\mathbf{P}}_k^- \mathbf{A} + \alpha_k \hat{\mathbf{P}}_k)^{-1} (\mathbf{A}^T \hat{\mathbf{P}}_k^- \mathbf{A} + \alpha_k \hat{\mathbf{P}}_k^-) \quad (36)$$

$$\hat{\mathbf{P}}_k = [\mathbf{I} - \bar{\mathbf{K}}_k \mathbf{H}] \hat{\mathbf{P}}_k^- \alpha_k^{-1} \quad (37)$$

In the above algorithm, $\bar{\mathbf{K}}_k$ is the modified robust adaptive Kalman gain, $\hat{\mathbf{x}}_k$ is the estimated state and $\hat{\mathbf{P}}_k$ is estimated state error covariance. $\mathbf{Q}_k$ is the estimated process noise covariance, $\hat{\mathbf{R}}_k$ is the estimated measurement noise covariance evaluated using equation (32) and α_k is the adaptive factor evaluated using the equation (30). The process noise covariance matrix $\hat{\mathbf{Q}}_k$ is estimated based on residual sequence, which is described in the appendix A. For more details, please ref [34].

VI. TEST RESULTS AND DISCUSSIONS

The experimental set up of IFOG sensor is shown in Fig. 1. The setup has a single axis IFOG sensor, DC-DC converter, data acquisition board and data processing computer. In this work, data is collected from a closed loop IFOG at room temperature. In the static condition, IFOG gyro is in zero degree rotation rate, while in the dynamic condition, the IFOG sensor is rotated with different rotation rate in the rate table.

A. Static Performance Test Analysis

Data is collected for 3 hrs duration with sampling frequency at 400 Hz for analysis and de-noising. Conventional Kalman filter algorithm is applied for de-noising the IFOG static signal. In CKF algorithm, the covariance measurement noise (R) and process noise (Q) are selected by trial and error method [23]. The CKF algorithm is applied for de-noising the IFOG signal under the quiescent condition, and depicted in Fig. 2. In real time, measurement (R) and process noise covariance (Q) matrices are varied with time. Hence, the CKF algorithm is not suitable for practical applications.

In IAE-AKF algorithm, the measurement noise covariance matrix $\hat{R}_k$ is estimated by using an innovation sequence by choosing window size as 15 samples [21], [23].

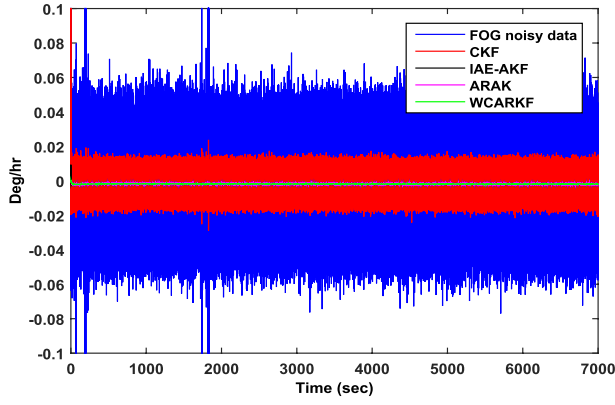


Fig. 2. x-axis FOG experimental results using proposed methods.

TABLE I

ALLAN VARIANCE RESULTS OF X-AXIS FOG SENSOR USING PROPOSED SCHEME IN STATIC CONDITION

	N (deg/ $\sqrt{\text{hr}}$)	Bs (deg/hr)	Bias drift (deg/hr)
FOG raw data	$2.40 \times 10^{(-2)}$	$2.2 \times 10^{(-1)}$	$1.54 \times 10^{(-1)}$
CKF	$2.09 \times 10^{(-2)}$	$1.8 \times 10^{(-2)}$	$2.0 \times 10^{(-2)}$
IAE-AKF	$1.50 \times 10^{(-3)}$	$1.3 \times 10^{(-2)}$	$2.5 \times 10^{(-3)}$
ARKF	$1.10 \times 10^{(-3)}$	$2.0 \times 10^{(-3)}$	$2.0 \times 10^{(-3)}$
WCARKF	$2.0 \times 10^{(-4)}$	$1.8 \times 10^{(-3)}$	$1.9 \times 10^{(-3)}$

In this work, window size was varied between 5 to 30 and observed that 15 as the optimal choice for statistical smoothing. The initial values of estimated state and state error covariance are assumed as zero and one respectively.

ARKF is also applied for filtering the IFOG signal under static condition (i.e., zero rotation rate). In this filter, an adaptive factor α_k is used for estimating the signal. The values of c_0 and c_1 are varied between 1.0 to 1.5 and 3.0 to 4.5 respectively. However, c_0 and c_1 are set as 1.2 and 4.5 respectively as the other choice was not able to follow the signal trend [29]. The de-noised signal using ARKF algorithm is also shown in Fig. 2. The Allan variance analysis results of the de-noised signal are tabulated in Table I.

The WCARKF algorithm is applied to IFOG signal for filtering the random noise. In this algorithm, the measurement noise covariance matrix is estimated by using the weighted covariance matrix of innovation sequence as (32). The weighted covariance is evaluated using (27) and (28), respectively. The adaptive factor is evaluated using (30) to update the estimated value of the signal. The performance of the proposed algorithm is compared with CKF, IAE-AKF, and ARKF as shown in Figs. 2 and 3. These Figures reveal that proposed algorithm outperform other algorithms.

In Fig. 3, the angle random walk (N) and bias instability are extracted from the slope -1/2 and zero respectively and tabulated in Table I. From this Table it is observed that angle random walk and bias instability are reduced by approximately ten times using the proposed algorithm. All the random noises are quantified before and after applying the algorithms. Bias drift is also calculated before and after de-noising the IFOG signal and tabulated in Table I. It is observed that the both

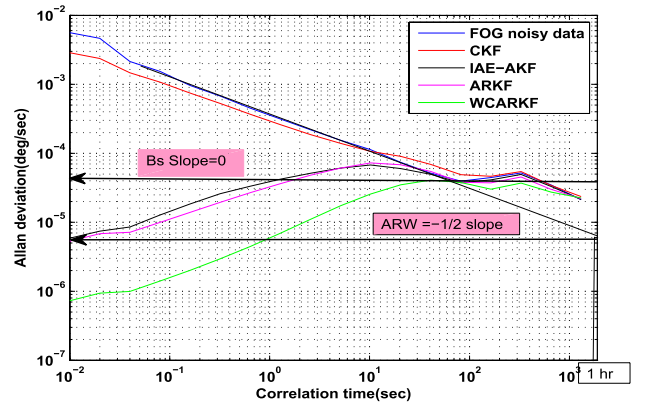
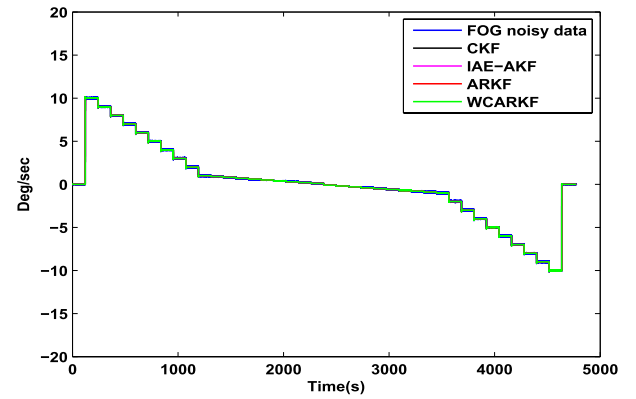


Fig. 3. x-axis FOG experimental results using Allan variance analysis plot using proposed methods.

Fig. 4. ± 10 deg/sec rotation FOG signal using denoised algorithms.

ARKF and WCARKF are performing better than conventional Kalman filter and IAE Kalman filter for minimizing random noise in IFOG signal.

B. Dynamic Performance Test Analysis

For acquiring the dynamic IFOG data, the rotation of the rate table is varied in a step manner starting from zero degree per second. In this study, ± 10 deg/sec step signals data are recorded at room temperature for 3 hrs duration with sampling frequency as 400 Hz.

For this data, the initial values of Kalman filter noise parameters R and Q are selected as 0.098 and 0.0001 respectively, using trial and error method.

Fig. 4 presents the noisy and de-noised signal obtained from the conventional KF (CKF), IAE-adaptive Kalman filter (IAE-AKF), adaptive robust Kalman filter (ARKF) and weighted covariance adaptive robust Kalman filter (WCARKF). These results correspond to the maneuvering (dynamic) test condition of IFOG. Fig. 5 presents the zoom version of Fig. 4 for the analysis purpose. From this Figure, it is observed that both ARKF and WCARKF outperforms CKF and IAE-AKF algorithm in terms of reducing noise and identifying discontinuities in the signal. Fig. 4 is further zoomed and shown in Fig. 6 to show the settling time of the algorithm to follow the signal. Fig. 6 reveals that WCARKF

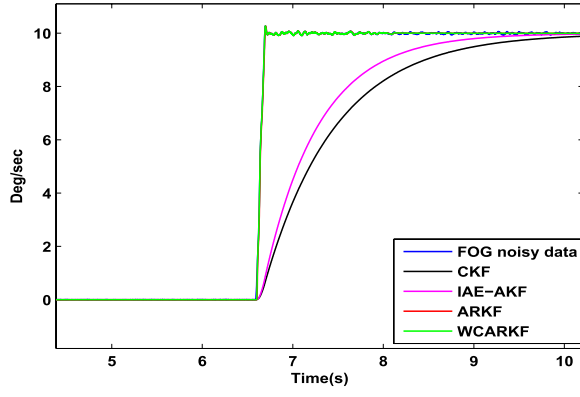
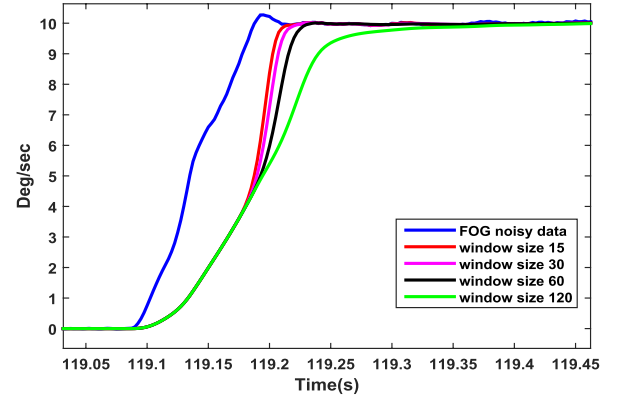
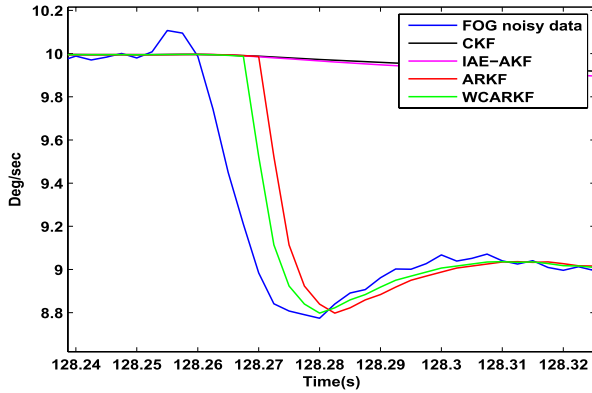
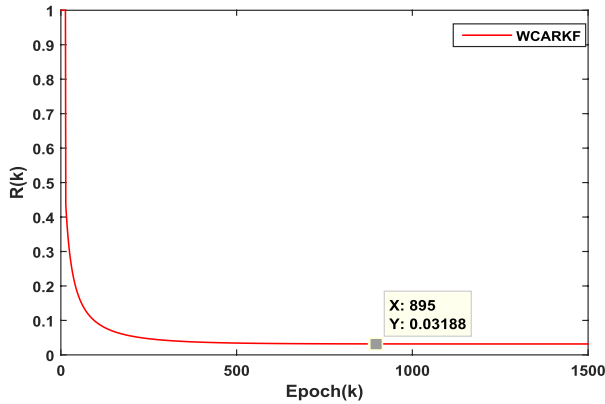
Fig. 5. ± 10 deg/sec step rotation FOG signal using denoised algorithms.

Fig. 8. Denoised signal for different window size using WCARKF algorithm.

Fig. 6. ± 10 deg/sec denoised signal with latency 0.01 second.Fig. 7. Adaption of R versus epoch using WCARKF algorithm.

and ARKF algorithm performs better than CKF and IAE AKF algorithm, in fact, WCARKF is slightly better than ARKF algorithm. Further we analysed the convergence rate of R and plotted in Fig. 7. It presents the variation of R with respect to epoch for the proposed algorithm. From this figure, it is observed that the algorithm takes approximately 895 epochs to converge to its stable value, i.e., 0.0318. We also analysed the impact of different window size on tracking the behavior of the WCARKF algorithm. Fig. 8 presents the step response of the algorithm for different window size. From this figure, it is observed that the optimum window size to settle the result

TABLE II
RMSE RESULTS OF IFOG SENSOR IN DYNAMIC CONDITION

	RMSE (deg/sec)
FOG raw data	4.80×10^{-2}
CKF	1.60×10^{-2}
IAE-AKF	1.40×10^{-2}
ARKF	1.30×10^{-2}
WCARKF	9.48×10^{-3}

to a stable value is 15. For larger window length, it takes more time to settle to the stable value.

The root mean squared error (RMSE) is used as the evaluation criteria to test the algorithm performance in dynamic environment. The RMSE is calculated before and after applying the algorithms to FOG signal as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (38)$$

where, $\hat{y}_i$ is the filtered signal, y_i is the raw IFOG data signal and n is the number of observations.

The RMSE of the signal for each step rotation of 50000 samples, i.e., 125 sec duration is calculated and tabulated in Table II. This table reveals that the WCARKF and ARKF algorithms are competitive. Although CKF and IAE-AKF algorithm also shows competitive performance, Fig. 5 concludes that CKF and IAE-AKF are not the good choices for minimizing the random noise of IFOG signal in maneuvering condition.

Based on the above experimental results, some important points are observed as follows. (1) From the Allan variance results and bias drift of the filtered signal as listed in Table I, it is observed that the random noise such as angle random walk and bias instability are reduced significantly in the case of WCARKF algorithm. The similar argument holds good for drift analysis. Thus, the modified robust adaptive Kalman filter is preferable for filtering random noise of IFOG in the static case.

(2) In the maneuvering case, both ARKF and WCARKF algorithm gives competitive results and outperforms both CKF and IAE AKF algorithm.

VII. CONCLUSION

In summary, the paper focused on the application of adaptive robust Kalman filter for suppressing random noise of IFOG signal in both static and maneuvering conditions. The robust adaptive Kalman filter is modified to weighted covariance adaptive robust Kalman filter to control the disturbances due to outlier measurements. In the modified algorithm, an adaptive tuning parameter and a weight function are used for adapting measurement noise covariance matrix R . In the static condition, the modified algorithm outperformed other algorithms as evident from Allan variance analysis and conventional bias drift analysis. Based on Allan variance analysis it is observed that the random noise of IFOG like angle random walk, bias instability and bias drift are reduced by a factor of hundred as compared to the true values. In the maneuvering case, both the adaptive robust Kalman filter and the modified algorithm gives competitive result but superior to conventional Kalman and IAE adaptive Kalman filter. Thus, it can be concluded that the modified algorithm namely weighted covariance adaptive robust Kalman filter is the suitable choice for noise minimization for IFOG signal. The hardware implementation of this algorithm in field programmable gate array is the future scope of this study.

APPENDIX

ADAPTION FOR ESTIMATING THE PROCESS NOISE
COVARIANCE MATRIX $\hat{Q}_k$

An estimated process noise covariance matrix $\hat{Q}_k$ can be defined based on residual sequence as

$$\hat{Q}_k = \frac{1}{D_s} \sum_{j=j_0}^k \omega(\sigma_{\hat{x}_{k-j}}^2) \Delta X_j \Delta X_j^T + \hat{P}_k - A \hat{P}_k A^T \quad (39)$$

$$\omega(\sigma_{\hat{x}_{k-j}}^2) = \left(\frac{1}{\hat{\sigma}_{\hat{x}_{k-j}}^2} \right) / \left(\sum_{j=1}^m \frac{1}{\hat{\sigma}_{\hat{x}_{k-j}}^2} \right) \quad (40)$$

where, $\hat{\sigma}_{\hat{x}_{k-j}}^2$ is the variance of predicted state vector defined as

$$\hat{\sigma}_{\hat{x}_{k-j}}^2 = \frac{\Delta X_k^T \hat{P}_k \Delta X_k}{z_k} \quad (41)$$

where, $\Delta X_k = \hat{x}_k - \hat{x}_k^-$ is the residual vector of the predicted state vector. Sum of weighted function of predicted state vector is equal to 1, i.e.,

$$\sum_{j=1}^m \omega(\hat{\sigma}_{\hat{x}_{k-j}}^2) = 1 \quad (42)$$

For more details on residual based adaptive estimation for process noise covariance matrix ($\hat{Q}_k$), please refer [34].

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